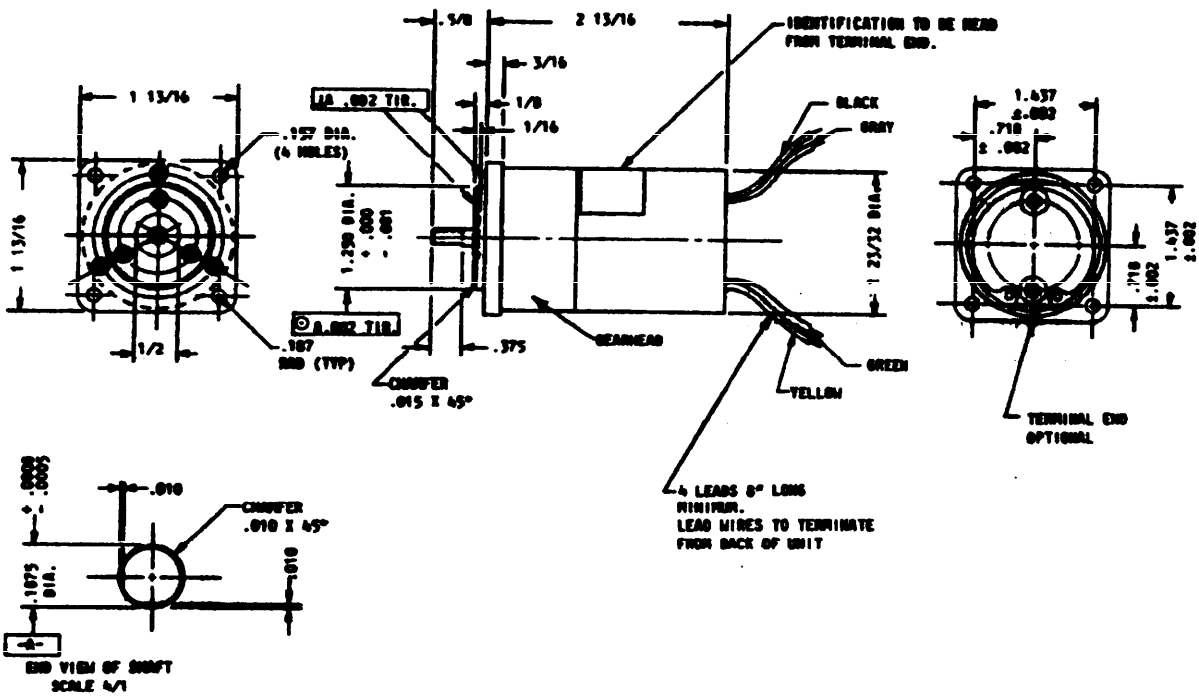
MILITARY SPECIFICATION SHEET
SERVOMOTOR TYPE 18SM6 (GEARHEADED)

THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR (GEARHEADED) DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-22432.



UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCE ON FRACTIONS ± 1/64
DECIMALS ± .005 ANGLES ± 5°

NOTES:

1. PRIMING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE GEARHEAD SERVOMOTORS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN MIL-S-22432.
2. DIMENSIONS AND TOLERANCES COMPLY WITH ASA B Y 14.5.
3. HOUSING MATERIAL SHALL MEET THE REQUIREMENTS OF 3.5.10 AND SHALL BE CAPABLE OF MEETING THE ENVIRONMENTAL REQUIREMENTS OF MIL-S-22432.

FIGURE 1

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TABLE I

SERVOMOTOR TYPE 18SM6 (GEARHEADED)

Performance Requirements and Characteristics

Requirement	Value	Unit	Tolerance
Motor (See Figure 2)			
Voltage (Test):			
Reference, (Fixed Phase)	115	volt	±1%
Control	115	volt	±1%
Frequency	60	Hz	±1%
Current (Stalled Condition):			
Reference (Fixed Phase)	40	milliamps	maximum
Control	40	milliamps	maximum
Power (Stalled Condition):			
Reference (Fixed Phase)	4	watts	maximum
Control	4	watts	maximum
Impedance (Stalled Condition):			
Reference (Fixed Phase)	3500	ohms	±20%
Control	3500	ohms	±20%
Power Factor (See Note 3):			
Reference (Fixed Phase)	.8	-	minimum
Control	.8	-	minimum
No Load Speed	3100	rpm	minimum
Minimum Starting Volts	2	volt	maximum
Stall Torque:			
25 Degrees C	.50	oz-in	minimum
50 Degrees C	.43	oz-in	minimum
D.C. Resistance (See Note 4):			
Reference (Fixed Phase)	1760	ohms	±15%
Control	1760	ohms	±15%

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TABLE I (Continued)

Other Requirements and Characteristics Motor With Gear Train (See Figure 3)

Requirement	Value	Unit	Tolerance
Speed (No Load)	1.7 - 2.2	rpm	min.-max.
Input Current (No Load)	65	milliamp	maximum
Input Watts (No Load)	7.5	watts	maximum
Minimum Starting Voltage (2.5 oz-in Load)	3	volt	maximum
Rotor Moment of Inertia	7.0	gm-cm^2	maximum
Temperature Rise (See Note 6)	50	degree	maximum
Weight	14	oz	maximum
Gear Ratio	1750-1		±.002
End Play (Gearhead Shaft) (See Note 5)	.008	inch	maximum
Radial Play (Gearhead Shaft) (See Note 5)	.002	inch	maximum

Direction of Rotation (See Note 7):

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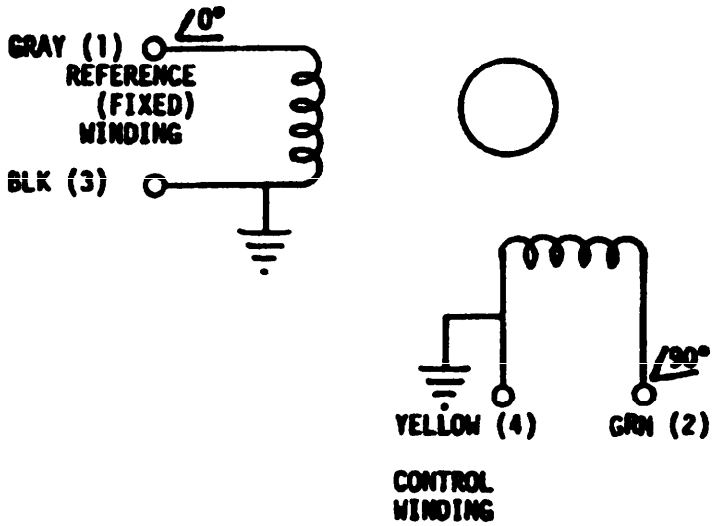


FIGURE 2

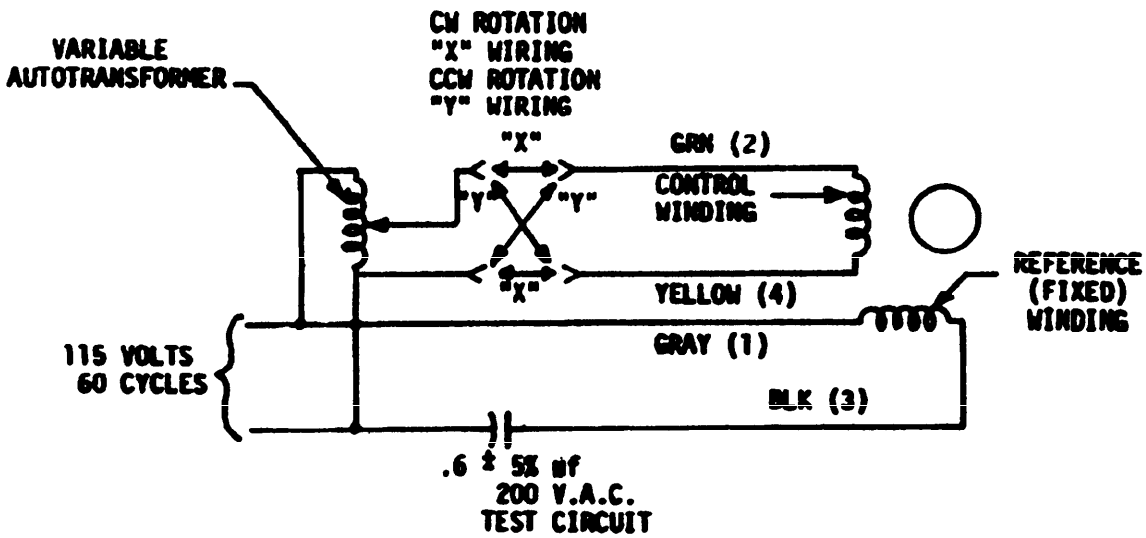


FIGURE 3

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NOTES:

1. The gearheaded servomotor identified herein shall be capable of operation in ambient temperatures ranging from -55 to 55 degrees C, and shall conform to all requirements and shall be subjected to all tests delineated in MIL-S-22432 with the following exceptions:

   Noise, Structure-Borne
   Salt Atmosphere Resistance
   Stall Torque Linearity
   Altitude

2. The following inspection shall be performed in addition to those listed in paragraph 4.8 of MIL-S-22432:

   "All encapsulated motors will be x-rayed to determine if there are any voids in the castings which might affect operation of the motor during projected life. See paragraph 3.33."

3. Power Factor - Measured by volt-ampere method.

4. D.C. resistance shall be measured at the ambient operating temperature of 25 degrees C ± 5 degrees C and in accordance with the remaining provisions of paragraph 4.2.3 of MIL-S-22432.

5. End Play and Radial Play shall be measured with reversal of the following loads:

   Radial - 8 oz.

   End - 16 oz.

6. Temperature rise shall be taken with the unit in stalled condition and mounted in a standard size 18 heat sink modified for mounting.

7. The direction of rotation shall be POSITIVE and CCW when the voltage at the YELLOW terminal wire electrically LEADS the voltage at the BLACK terminal wire.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 6105-A-072

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